

Global Facilities - Mechanical – Process Exhaust System Standard (AE, AME, SE, GE)

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1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Process Exhaust Systems as part of new Greenfield FAB Construction projects.
- 1.2 The standard is applicable for Acid Exhaust, Ammonia Exhaust, General Exhaust and Solvent Exhaust (AE, AME, GE, SE) systems.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 This document covers the new Basebuild System installations only, as part of new Greenfield FAB Construction projects. This document does NOT cover Tool Installation or Point-of-Use (POU) Abatement.
- 2.2 This document does not cover other Exhaust systems, such as Non-Process or General Building (GHVAC) Exhaust, Smoke Exhaust, or other Specialized Exhaust systems.
- 2.3 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
AE	Acid Exhaust
AHJ	Authority having Jurisdiction
AME	Ammonia Exhaust (also known as Alkali Exhaust or Caustic Exhaust)
FMCS/SCADA	Facility Monitoring and Control System / Supervisory Control and Data Acquisition
GE	General Exhaust
GHVAC	General Heating, Ventilation, Air Conditioning
HAP	Hazardous Air Pollutants
LCP	Local Control Panel
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
N+1+F	System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, + future unit(s)
PLC	Programmable Logic Controller
POC/LPOC	Point of Connection / Lateral Point of Connection
POU	Point of Use
SE	Solvent Exhaust (also known as VOC Exhaust or Organic Exhaust)

Term/Acronym	Definition / Description
TUM	Tool Utility Matrix
UPS	Uninterruptible Power Supply
UPW/DI/RO	Ultrapure Water / De-Ionized Water / Reverse Osmosis Water
VMC	Valve Manifold Box
VOC	Volatile Organic Compounds
VSD/VFD	Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

	Responsibilities
GFTT Lead (Discipline Engineer)	- Develop and maintain System Standard document - Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard - Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	- Review and provide technical feedback on System Standard - Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	- Review and provide technical feedback on System Standard - Incorporate System Standard into Project Engineering, Design, and Construction - Document project deviations, exceptions, and exclusions from System Standard - Share new lessons learned to GFTT lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	Use System Standard as technical reference

5.0 Overview of Systems

The following Exhaust systems are used to support the requirements from Manufacturing and Facilities loads:

- 5.1 **Acid exhaust (AE)** is used to remove and neutralize acidic compounds from tool process exhaust via abatement scrubbers, before discharging the exhaust to atmosphere as clean air.

The hazardous air pollutants (HAP) that are abated by the Acid Exhaust system include Chlorine (Cl), Hydrochloric acid (HCl), Hydrofluoric acid (HF), Phosphoric acid (H₃PO₄), Hydrobromic acid (HBr), Nitric acid (HNO₃), Sulfuric acid (H₂SO₄), and Acetic acid (CH₃COOH).

The main AE Exhaust System equipment shall be as follows:

- 5.1.1 Exhaust Scrubbers
- 5.1.2 Exhaust Fans
- 5.1.3 Exhaust Stacks
- 5.1.4 Chemical dosing system

- 5.2 **Ammonia exhaust (AME)** is used to remove and neutralize ammonia (NH₃, NH₄) compounds from tool process exhaust via abatement scrubbers, before discharging the exhaust to atmosphere as clean air. Ammonia Exhaust is also referred to as Alkali Exhaust or Caustic Exhaust.

The main AME Exhaust System equipment shall be as follows:

- 5.2.1 Exhaust Scrubbers
- 5.2.2 Exhaust Fans
- 5.2.3 Exhaust Stacks
- 5.2.4 Chemical dosing system

- 5.3 **Solvent exhaust (SE)** is used to remove and destroy Volatile Organic Compounds (VOC) from tool process exhaust via VOC abatement units, before discharging the exhaust to atmosphere via clean air. The typical VOCs include Isopropyl Alcohol (IPA, C₃H₈O), Methyl Propyl Acetate (PGMEA, C₆H₁₂O₂), Butane (C₄H₁₀), and Benzene (C₆H₆). Solvent Exhaust is also referred to as VOC Exhaust or Organic Exhaust.

The main Process SE Exhaust System equipment shall be as follows:

- 5.3.1 VOC Abatement Units
- 5.3.2 Exhaust Fans
- 5.3.3 Exhaust Stacks

Note: VOC Abatement units shall be installed for all Solvent Exhaust systems, regardless of whether or not it is required by local/regional/national code, Authority Having Jurisdiction (AHJ), or air emissions regulations.

- 5.4 **General Exhaust (GE)** is used to remove non-corrosive, non-toxic, non-hazardous, or heated exhaust from tool process exhaust and then discharging the exhaust directly to atmosphere without any processing or abatement.

The main Process GE Exhaust System equipment shall be as follows:

5.4.1 Exhaust Fans

5.4.2 Exhaust Stacks

The typical uses for General Exhaust include heat exhaust, cabinet enclosure exhaust, and chamber exhaust from tools including Diffusion, FAB Support equipment.

6.0 Performance Specs

- 6.1 The Process Exhaust Systems shall be designed and installed to achieve the operational performance specs as outlined below; See **Table 1**.

Table 1: Operational Performance Specs for Exhaust Systems

Parameter	Specification Requirement	Remarks
Acid Exhaust Pressure	-900 ± 50 Pa (-3.6 ± 0.2 in W.C.)	Measured at end-of- line in Subfab level lateral.
Ammonia Exhaust Pressure	-900 ± 50 Pa (-3.6 ± 0.2 in W.C.)	Measured at end-of- line in Subfab level lateral.
Solvent Exhaust Pressure	-900 ± 50 Pa (-3.6 ± 0.2 in W.C.)	Measured at end-of- line in Subfab level lateral.
General Exhaust Pressure	-750 ± 50 Pa (-3.0 ± 0.2 in W.C.)	Measured at end-of- line in Subfab level lateral.

Note: The Exhaust Systems shall be designed and installed with capability to meet the exhaust pressures stated above. After installation, the actual operation of the systems and setpoints for end-of-line pressures for each lateral may be different (less) than the pressures shown. This is due to different pressure requirements for different Process Areas and Tool Sets.

- 6.2 The Process Exhaust System Equipment shall be designed and selected to achieve the minimum removal efficiencies as outlined below; See **Table 2**.

Table 2: Removal Efficiency Performance Specs for Exhaust Systems

Parameter	Specification Requirement
Acid Exhaust (AE)	- Minimum Removal efficiency of 95% for the following compounds: - Hydrofluoric Acid (HF) - Hydrochloric Acid (HCl) - Sulfuric Acid (H₂SO₄) - Nitric Acid (HNO₃)
Ammonia Exhaust (AME)	- Minimum Removal efficiency of 95% for the following compounds: - Ammonia (NH₃)

Solvent Exhaust (SE)	Minimum Destruction Removal Efficiency (DRE) of 95% overall for VOCs
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Note: If a higher removal efficiency is required by the local/regional/national code, AHJ, and /or air emissions regulations, than the stricter requirement shall take precedence and shall be met.

7.0 Equipment Sizing & Design Requirements:

7.1 Redundancy & Future – Exhaust System main equipment shall be designed for N+1+F or N+1+2F for Scrubbers/Abatement Units and Fans. All main system equipment shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install additional future units (F or 2F) for future capacity expansion.

7.1.1 Acid Exhaust: Acid Exhaust System main equipment shall be designed for 2 future units for scrubbers and fans; N+1+2F.

7.1.2 Ammonia Exhaust: Ammonia Exhaust System main equipment shall be designed for 1 future unit for scrubbers and fans; N+1+F.

7.1.3 General Exhaust: General Exhaust System main equipment shall be designed for 2 future units for fans; N+1+2F.

7.1.4 Solvent Exhaust: Solvent Exhaust System main equipment shall be designed for 2 future units for abatement units and fans; N+1+2F.

7.1.5 Each Acid and Ammonia Exhaust Scrubber shall have N+1 recirculation pump redundancy for the individual Scrubber.

7.1.6 Each Acid and Ammonia Exhaust Chemical Dosing unit shall have N+1 chemical supply pumps for each individual unit.

7.2 System Diversity and Safety Factor – Exhaust System sizing shall be selected so that the N capacity meets 115-120% of the exhaust demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM, plus the Facilities Basebuild requirements. Therefore, the Exhaust system sizing capacity for N capacity shall have 15-20% spare future capacity over the exhaust demand at full build (after initial construction of the system).

7.2.1 To determine the Exhaust demand at N full build, a diversity factor of 100% of the Peak demand from the TUM shall be

used. In addition, the exhaust system shall be sized to include the capacity of future units (F or 2F).

- Acid, General, and Solvent Exhaust Systems shall be designed to have minimum 2 Future units (N+2F minimum).
- Ammonia Exhaust Systems shall be designed to have minimum 1 Future unit (N+1F minimum).

7.2.2 The Facilities Basebuild requirements shall include all other users of exhaust such as Gas Cabinets, Gas and Chemical VMBs, Chemical Supply Equipment, UPW and Wastewater Tanks and Equipment, Room Area exhaust, etc.

7.2.3 See **Tables 3 and 4** for Exhaust system sizing criteria and calculation.

Table 3: System Sizing Calculation – Acid, General, Solvent Exhaust

- Exhaust System Demand at Full Build, D:

$$D = (\text{TUM Peak Demand} \times 100\% \text{ Diversity}) + \text{Facilities Basebuild Requirements}^*$$
- Exhaust System N capacity:

$$N = 120\% \times D$$
- Exhaust System Sizing to include future units:

$$\text{Sizing} = N + 2F$$

Table 4: System Sizing Calculation – Ammonia Exhaust

- Exhaust System Demand at Full Build, D:

$$D = (\text{TUM Peak Demand} \times 100\% \text{ Diversity}) + \text{Facilities Basebuild Requirements}^*$$
- Exhaust System N capacity:

$$N = 120\% \times D$$
- Exhaust System Sizing to include future unit:

$$\text{Sizing} = N + F$$

- 7.3 **Equipment Quantity** – Exhaust System main equipment shall be selected for a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing; for scrubbers/abatement units and fans. The exhaust demand at full build shall be split evenly across the number of equally sized exhaust equipment.
- 7.4 **Equipment Arrangement, Acid and Ammonia Exhaust**– All Main equipment shall be arranged in parallel together and connected via common duct headers upstream and downstream of the scrubbers. This allows any combination of equipment to be in operation concurrently and allow for redundancy on all equipment.
- 7.4.1 Exhaust Scrubbers shall be connected to a common ductwork header upstream of the scrubbers. The header shall be constant diameter size across the entire length of the header.
- There shall be a multi-blade type motorized control damper and manual blastgate on the inlet ductwork to the Scrubber. The control damper shall be programmed to open upon scrubber startup, and close during scrubber shutdown or fault.
 - There shall be a multi-blade type manual damper and manual blastgate on the discharge ductwork from the Scrubber.
- 7.4.2 Exhaust Fans shall be connected to a common ductwork header upstream of the fans and downstream of the scrubbers. The header shall be constant diameter size across the entire length of the header.
- There shall multi-blade type motorized control damper and manual blastgate on the inlet ductwork to the fan. The control damper shall be programmed to open upon fan startup, and close during fan shutdown or fault.
 - Each fan shall be connected to a dedicated discharge stack.
 - It is recommended to keep a minimum of 3- to 5-duct diameters of unobstructed straight duct upon entrance into the exhaust fan inlet, to prevent any fan system effects or pre-swirling and turbulence.
- 7.5 **Equipment Arrangement, Solvent Exhaust**– All Main equipment shall be arranged in parallel together and connected via common duct headers upstream and downstream of the abatement units. This allows any combination of equipment to be in operation concurrently and allow for redundancy on all equipment.

- 7.5.1 VOC Abatement Units shall be connected to a common ductwork header upstream of the units. The header shall be constant diameter size across the entire length of the header.
- There shall be a multi-blade type motorized control damper and manual blastgate on the inlet ductwork to the Abatement Unit. The control damper shall be programmed to open upon VOC Abatement Unit startup, and close during unit shutdown or fault.
 - There shall be a multi-blade type motorized control damper and manual blastgate on the discharge ductwork from the Abatement Unit. The control damper shall be programmed to open upon VOC Abatement Unit startup, and close during unit shutdown or fault.
- 7.5.2 Exhaust Fans shall be connected to a common ductwork header upstream of the fans and downstream of the Abatement Units. The header shall be constant diameter size across the entire length of the header.
- There shall multi-blade type motorized control damper and manual blastgate on the inlet ductwork to the fan. The control damper shall be programmed to open upon fan startup, and close during fan shutdown or fault.
 - Each fan shall be connected to a dedicated discharge stack.
 - It is recommended to keep a minimum of 3- to 5-duct diameters of unobstructed straight duct upon entrance into the exhaust fan inlet, to prevent any fan system effects or pre-swirling and turbulence.
- 7.5.3 There shall be a main bypass duct with multi-blade type bypass damper, in between the VOC Abatement Unit header and the Exhaust Fan header. The bypass shall be used when a VOC Unit is offline or during startup/shutdown transitions of VOC units.
- The bypass damper can be either motorized, modulating type control damper or gravity backdraft damper type. If motorized type, the control damper shall be programmed to open upon shutdown or fault of VOC Unit, and close during startup or during operation of VOC Unit.
 - The bypass damper shall be adjusted and tuned to minimize pressure fluctuations during operation.
- 7.5.4 Exhaust Fans configuration for VOC Abatement Unit shall be Primary-Secondary Fan configuration, to overcome the pressure drop of the VOC Abatement Unit.
- The Primary Exhaust fans shall be located upstream of the VOC Abatement Units, and the Secondary Exhaust fans shall be located downstream of the VOC Abatement Units.
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- 7.6 **Equipment Arrangement, General Exhaust** – All Main equipment shall be arranged in parallel together and connected via a common duct header. This allows any combination of equipment to be in operation concurrently and allow for redundancy on all equipment.
- 7.6.1 Exhaust Fans shall be connected to a common ductwork header upstream of the fans. The header shall be constant diameter size across the entire length of the header.
- There shall multi-blade type motorized control damper and manual blastgate on the inlet ductwork to the fan. The control damper shall be programmed to open upon fan startup, and close during fan shutdown or fault.
 - Each fan shall be connected to a dedicated discharge stack.
 - It is recommended to keep a minimum of 3- to 5-duct diameters of unobstructed straight duct upon entrance into the exhaust fan inlet, to prevent any fan system effects or pre-swirling and turbulence.
- 7.7 **Equipment Layout** – Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment, and to allow move-in and move-out of large components during maintenance and repair.
- 7.7.1 Exhaust Fans shall have adequate clearance space to allow maintenance, repairs, and/or replacement on fan assembly, impeller, and motor.
- 7.7.2 Exhaust Scrubbers shall have adequate clearance space to allow maintenance, repairs, and/or replacement on scrubber pumps, instrumentation, packing, spray headers, and mist eliminators.
- 7.7.3 VOC Abatement Units shall have adequate clearance space to allow maintenance, repairs, and/or replacement on zeolite rotors, gas train assembly, concentrator chamber, oxidizer chamber, all integral fans, dampers, instrumentation, etc.
- 7.8 **Equipment Location, Exhaust Fans** – It is recommended to locate and install Exhaust Fans at an upper floor level, if possible. This is to reduce the length required for the fan discharge stack. Also, it is recommended to locate the Exhaust Fans indoors, if possible. This is to allow easier operations and maintenance activities on the fans, and eliminate the requirements for weatherproofing.
- 7.9 **Equipment Location, Exhaust Scrubbers** – It is recommended to locate and install Exhaust Scrubbers indoors, if possible. This is to allow easier operations and maintenance activities on the scrubbers, eliminate

the requirements for weatherproofing, and prevent potential freezing of components in locations with cold climates.

7.9.1 If located outdoors, provide proper weatherproofing covers, weather protection, and freeze protection for all critical instrumentation, devices, and components.

7.10 **Equipment Location, VOC Abatement Units** – It is recommended to locate and install VOC Abatement units outdoors, if possible. This is to eliminate the amount of heat gain to the space, room ventilation requirements, and room fire protection requirements.

7.10.1 If located outdoors, provide proper weatherproofing covers, weather protection, and freeze protection for all critical instrumentation, devices, and components.

7.11 **Equipment Location, Exhaust Discharge Stacks** – Exhaust stacks shall be located away from Makeup Air Unit (MAU) intakes so that air discharged from the stacks is not re-entrained into the MAU intakes.

7.11.1 Exhaust Discharge Stack height shall comply with local/regional/national code, AHJ, and/or air emissions regulations requirements.

7.11.2 Computational Fluid Dynamics (CFD) Modeling shall be performed to simulate the path of exhaust discharge from the stacks. The modeling shall be used to ensure that exhaust discharge is not re-entrained into the MAU intakes.

7.12 **Pressure Drop Calculations** – Submit pressure drop calculations for each ductwork system, to show that each fan selection and the fan parameters (flow, pressure, power, etc.) are adequate to meet the design requirements.

7.13 **Equipment Selection Criteria** – See **Tables 5 to 7** for Exhaust system selection criteria and requirements.

Table 5: Exhaust Fan and Discharge Stack Equipment Selection Criteria – Acid, Ammonia, Solvent, and General Exhaust

Equipment	Type & Description
Exhaust Fans – All (Acid, Ammonia, Solvent, and General Exhaust)	- Exhaust Fans shall be Centrifugal type, Direct-Drive type. - Exhaust Fan sizing and selection shall be as follows: - Fans shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower).

	○ Fan curve shall show the performance indicated at the maximum operating point within the fan motor size limitations (flow and pressure, based on fan impeller size, VSD speed, and full brake-horsepower to match motor size rating). Resultant fan tip speed shall be stated. ○ The maximum allowable fan impeller tip speed rating shall be published. ○ The fan maximum operating point tip speed shall be at least 15% - 20% less than the maximum allowable tip speed of the fan. Meaning, for continuous operation at the fan maximum operating point, the fan impeller tip speed shall be less than 80% - 85% of the maximum allowable fan impeller tip speed rating. ○ Fan impeller shall be AMCA rated, Class IV ○ Heavy duty bearings ○ Shaft shall have critical speed 125% of fans maximum operating speed ○ Mechanical efficiency shall be 75% above. ○ Fan assembly shall be statically and dynamically balanced. ○ Totally enclosed fan-cooled motor with over-temperature protection. ○ Sound level performance to be < 75 dBA at distance of 1m away from fan enclosure. • Fan motor sizing and selection shall be as follows: ○ Fan motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions. ○ Fan motors shall be rated for inverter duty with variable speed operation. ○ Fan motor efficiency rating shall be minimum IE3. ○ Fan motor IP rating shall be minimum IP55. If located outdoors, fan motor IP rating shall be IP56. • Fan installation shall be as follows: ○ Each Exhaust Fan shall have concrete inertia base and vibration isolation springs. ○ Each Exhaust Fan housing shall have access door to allow inspection and cleaning of fan impeller.
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	○ Each Exhaust Fan housing shall have drain connection port. ○ Each Exhaust Fan shall have drain piping connected from the drain connection port. ○ Each exhaust fan shall have minimum 25mm (1-inch) test ports upstream and downstream of the fan to allow airflow and pressure readings. ○ Each exhaust fan shall have minimum 150mm (6-inches) cleaning port upstream and downstream of the fan to allow inspection and cleaning. ○ Each Fan shall have flexible ductwork connection on inlet side and flexible ductwork connection on discharge side. <u>Acid and Ammonia Exhaust Fans</u> ○ Fan housing shall be all Fiberglass materials of construction. ○ Fan impeller shall be made of Fiberglass, Carbon Fiber, or hybrid combination of Fiberglass and Carbon Fiber. Note: Fan impellers made of Steel, Rubber-lined Steel, or Teflon-lined Steel materials of construction shall NOT be used. Note: Approved Manufacturers – Verantis, Hartzell, Winfan, Kyowa, or approved equivalent <u>Solvent Exhaust Fans</u> ○ Fan housing shall be hot dipped galvanized Steel materials of construction. ○ Fan impeller shall be made of hot dipped galvanized Steel. ○ Fan be of AMCA Class A Spark Resistant Construction for all parts in contact with airstream (fan impeller, shaft, etc.) in accordance with ANSI/AMCA Standard 99-16 and AMCA Standard 99-0401-86 Note: Approved Manufacturers – New York Blower (NYB), Twin City Fan (TCF), or approved equivalent <u>General Exhaust Fans</u>
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	○ Fan housing shall be hot dipped galvanized Steel materials of construction. ○ Fan impeller shall be AMCA Class IV rated. ○ Fan impeller shall be made of hot dipped galvanized Steel. ○ It is optional for fan be of AMCA Spark Resistant Construction. Note: Approved Manufacturers – New York Blower (NYB), Twin City Fan (TCF), or approved equivalent
Exhaust Discharge Stacks - All (Acid, Ammonia, Solvent, and General Exhaust)	• Exhaust Discharge Stacks shall be free-standing design. • Exhaust Discharge Stack height shall comply with local/regional/national code, AHJ, and/or air emission regulations requirements. • Exhaust Discharge Stacks shall be designed to withstand maximum wind load of 100 km/hr or maximum wind speed condition at the site location. • Stack velocity shall be designed for minimum 3000-4000 fpm (15-20 m/s) at full load • Provide Discharge Stacks with the following: ○ Flanged inlet connection ○ Lightning protection accessories and provisions ○ Access door ○ Drain connection at bottom of stack and drain system to prevent rain water from entering into duct or fan system ○ Sampling ports ○ Access platforms, ladders, and stairways as required to perform regular maintenance and emissions testing at sampling port locations <u>Acid and Ammonia Exhaust - Discharge Stacks</u> ○ Discharge Stack shall be Fiberglass Reinforced Plastic or Coated Stainless Steel materials of construction. <u>General and Solvent Exhaust - Discharge Stacks</u> ○ Discharge Stack shall be Stainless Steel materials of construction.

Table 6: Exhaust Scrubber Equipment Selection Criteria – Acid and Ammonia Exhaust

Exhaust Scrubbers - Acid Exhaust	• Scrubbers shall be Vertical counter-current packed-bed Wet scrubbers, or Horizontal cross-flow packed bed Wet scrubbers
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and Ammonia Exhaust

- For Vertical counter-current type scrubbers, it is recommended to use cylindrical shape scrubber bodies instead of rectangular shape scrubber bodies. This is for improved uniform air distribution and prevent bypassing of air around the packing.
- Scrubber shall consist of the following components:
 - Housing
 - Sump
 - Packing and associated supports
 - Spray Nozzles and associated piping and supports
 - Mist Eliminator and associated supports
 - Minimum 2 Recirculation Pumps (N+1, with 1 duty and 1 offline standby)
 - Access Manholes and Ports
 - Overflow Drain
 - Double Containment Pan with Drain and Leak Detection
 - All piping, valves, accessories, and supports
 - All wiring, cables and associated supports for Electrical Power, Instrumentation and Controls, etc.
 - All sensors, transmitters, switches, and gauges
 - Local Control Panel including PLC, HMI, etc.

Scrubber Body and Shell

- Scrubber body and shell shall be made of Fiberglass Reinforced Plastic (FRP) material, fire retardant grade.

Mist Eliminator

- Mist Eliminator shall be Chevron type or Packing type.

Note: Mesh pad type mist eliminators shall NOT be used.

- Mist eliminator housings shall be designed to drain back into the scrubber sump with the capability of removing 99% of the entrained moisture in the airstream.
- Mist eliminators shall have an effective removal of liquid droplets down to 10 microns.
- If wide-angle spray nozzles are used on the recirculation water spray distribution header, there shall be a minimum distance of 2ft. (0.6m) between the distribution header and the top of the packing to eliminate water spray

	bouncing off the packing and overloading the mist eliminator. <u>Sump</u> ○ A scrubber sump shall be integral to the body of the scrubber and provided with a water-tight lid with a clear polycarbonate viewport. ○ The bottom of a scrubber sump shall be one-piece integral to the scrubber housing, reinforced to support the weight of the scrubber plus the normal liquid charge and the weight of service personnel. ○ Sump to have a minimum capacity of 110 percent of the maximum volume of solution in circulation at any time. Maximum solution level not to exceed the bottom of the inlet duct. <u>Recirculation Pumps and Piping</u> ○ Recirculation water piping shall be sized for a maximum velocity not to exceed 5 ft./second (1.5 m/second). ○ Recirculation water pumps shall be made of FRP or PP materials of construction, or other chemically compatible materials. ○ Pump motor IP rating shall be minimum IP55. If located outdoors, pump motor IP rating shall be IP56. ○ Provide pumps and recirculation piping with the following: - Flexible connections that are compatible with water chemistry. - Strainer - Discharge check valve at the pump outlet - Pump vibration isolation, if required ○ It is recommended to connect the Scrubber chemical dosing injection directly to the Scrubber recirculation piping, for improved mixing and pH control. ○ Provide isolation valves at both sides of recirculation piping flow transmitter, and bypass piping with isolation valve around the flow transmitter, to allow isolation for maintenance, repair, or calibration ○ Provide isolation valves at both sides of makeup water control valve and bypass piping with isolation valve around makeup water control valve, to allow isolation for maintenance or repair
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	○ Provide isolation valves at both sides of blowdown drain control valve, and bypass piping with isolation valve around the blowdown drain control valve, to allow isolation for maintenance or repair • Scrubber sizing and selection shall be as follows: ○ Packing depth: 5 ft. (1.5 m) minimum ○ Packing face velocity: 500 fpm (2.54 m/s) maximum ○ Typical packing pressure drop: 0.25 in. W.C. (62.2 Pa) per foot of packing depth, at design air flowrate and clean packing conditions ○ Typical mist eliminator face velocity: 750 fpm (229 m/min) at design air flowrate. • Scrubber shall have the following Instrumentation transmitters, with readouts to the Local Control Panel and to SCADA ○ Recirculation Flowmeter ○ pH Meter ○ Conductivity Meter ○ Packing differential pressure transmitter ○ Sump Level Sensor ○ Double Containment Pan Leak Detection ○ Makeup water valve position and status ○ Blowdown valve position and status ○ Chemical dosing valve position and status ○ Provide pH and conductivity elements that can be mechanically isolated to allow probe calibration while scrubber is in operation. • Scrubber shall have following manual gauges with local readouts: ○ Recirculation Pump discharge pressure ○ Sump level sight glass. ○ Packing differential pressure gauge ○ Demister differential pressure gauge ○ Makeup water flowmeter ○ Blowdown flowmeter • Scrubber installation shall be as follows: ○ Each Scrubber shall have PVC flexible ductwork connection on inlet side and discharge side of scrubber. ○ Each Scrubber shall have sprinkler head installed on ductwork at inlet side and discharge side of scrubber.
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	○ Each Scrubber shall have minimum 25mm (1-inch) test ports upstream and downstream of the scrubber, to allow airflow, pressure, and air quality measurements. ○ Inlet ductwork to scrubber shall be designed so that any liquid or condensation in the exhaust ductwork will drain back into scrubber sump. The scrubber inlet elevation shall be at a lower height than the exhaust duct header elevation. ○ Provide access platforms, ladders, and stairways as needed to perform regular maintenance and repair/replacement on scrubber components including packing, spray nozzles, and demister. ○ Scrubber drain piping routed along the floor shall not obstruct or block any corridors or main walkways, for walking or cart access Note: Approved Manufacturers – Verantis, Viron, or approved equivalent
Chemical Dosing Skid, Acid and Ammonia Exhaust	• Provide a NaOH (Sodium Hydroxide) chemical dosing system for pH adjustment for the Acid Exhaust Scrubbers, consisting of Centralized Bulk Tank and Localized Day Tanks, supply pumps, and distribution system • Provide a central H2SO4 (Sulfuric Acid) chemical dosing system for pH adjustment for the Ammonia Exhaust Scrubbers, consisting of Centralized Bulk Tank and Localized Day Tanks, supply pumps, and distribution system • It is recommended to connect the Scrubber chemical dosing injection directly to the Scrubber recirculation piping, for improved mixing and pH control. • The Centralized Bulk chemical dosing systems shall be used to fill the localized day tanks, and shall consist of the following: ○ Two bulk chemical tanks with an auto-crossover system, equipped with valve manifold to allow isolating one side for maintenance while supplying from the other side in order to supply chemical continuously ○ Chemical supply pumps with redundant backup (N+1), with isolation valves, pressure gauges, check valves, etc. ○ Tank Level sensors and transmitters ○ Tank Level sight glass

	○ Double contained chemical distribution piping from bulk chemical tanks to localized day tanks ○ Vented exhaust ductwork to appropriate exhaust system ○ Local control panel with PLC, manual override controls, indicator lights, status monitoring, alarms, emergency off button (EMO), etc. ○ Enclosure cover around chemical supply pumps and discharge piping/fittings to contain splashes, spills, and leaks ○ Double Containment Pan with Leak Detection • The Localized Day Tank chemical dosing systems shall be used to supply chemicals to each Scrubber to maintain Scrubber pH control. There can be one Localized Day Tank system serving each scrubber, or one Localized Day Tank system serving multiple or all scrubbers. The Localized Day Tank chemical dosing system shall consist of the following: ○ Day tank with minimum 48-hour chemical storage capacity ○ Chemical supply pumps with redundant backup (N+1), with isolation valves, pressure gauges, check valves, etc. ○ Tank Level sensors and transmitters ○ Tank Level sight glass ○ Double contained chemical distribution piping from day tank to Scrubbers, with open/close type chemical dosing control valve at each Scrubber ○ Vented exhaust ductwork to appropriate exhaust system ○ Local control panel with PLC, manual override controls, indicator lights, status monitoring, alarms, etc. ○ Enclosure cover around chemical supply pumps and discharge piping/fittings to contain splashes, spills, and leaks ○ Double Containment Pan with Leak Detection
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Table 7: Solvent Exhaust Equipment Selection Criteria

Solvent Exhaust – VOC Abatement system	• VOC Abatement Unit shall be Rotary Concentrator Thermal Oxidizer type. • VOC Abatement Unit shall be provided as complete system assembly, including but not limited to the following integral components: ○ Zeolite Rotor assembly(s) ○ Concentrator assembly ○ Thermal Oxidizer assembly ○ Cooling/Desorption Fan ○ Oxidizer/Concentrate Fan ○ Combustion Air Fan ○ All Instrumentation and Controls devices including Dampers, Valves, associated actuators and position switches, etc. ○ Main Inlet and Outlet Damper ○ All interconnecting piping and ductwork, associated supports, and accessories for Process Exhaust, Natural Gas, Instrument Air, Fire Suppression, Nitrogen, Drains, etc. ○ All interconnecting wiring and cables and associated supports for Electrical Power, Instrumentation and Controls, etc. ○ All temperature and pressure sensors, transmitters, switches, and gauges ○ All filter assemblies including Pre-filter, Combustion Air filter, etc. ○ Local Control Panel including PLC, HMI, etc. PLC Controls shall have redundancy including redundant CPU modules, dual power source, Remote I/O isolation, and hot swap capability. ○ Variable frequency Drives (VFD) for rotor(s) and/or integral fans, if applicable ○ Access doors and view windows ○ Interior lighting for accessible sections, and external lighting if applicable ○ Platforms, ladders, and stairways required for regular system maintenance and testing access. ○ Natural gas and fuel train assembly consisting of valves, pressure regulators, gauges, filter/strainer, safety valves, switches, flame rods, and burner assembly ○ All unit safety controls and interlocks including burner assembly, etc. ○ All unit operational status monitoring and alarm notification
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	• VOC Abatement Unit shall have single POC for all utility supply connections, including but not limited to the following: ○ Inlet and outlet Exhaust ductwork. ○ Electrical Power ○ Instrument air (IA). ○ Nitrogen (N2). ○ Waste Drainage. ○ Natural Gas (NG) ○ Fire protection and sprinkler piping. ○ Instrumentation and Controls • VOC Abatement Unit installation shall be as follows: ○ Provide natural gas piping to VOC Abatement Unit including dual pressure regulators, isolation valve, bypass piping around pressure regulators, pressure gauges, and flowmeter Note: Approved Manufacturers – Munters, Durr, or approved equivalent
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8.0 Duct Design and Sizing Requirements:

8.1 **Ductwork Diversity and Sizing Factor** – The Exhaust system ductwork shall be sized according to the same sizing factors (N+F, System Diversity, etc.) stated in the above section 7.2.3 and below.

8.2 **Ductwork Mains Sizing Calculation** – Exhaust Mains shall be sized to accommodate 115-120% of the Exhaust demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM, plus the Facilities Basebuild Requirements. Therefore, the Exhaust Mains shall have 15-20% spare future capacity over the Exhaust demand at full build after initial construction of the system). In addition, the Exhaust Mains shall be sized to include the capacity of future units (F or 2F). See **Table 8**.

Because the Exhaust system equipment shall be installed in parallel configuration, the exhaust load shall be split evenly across the Exhaust Main ductwork, and therefore the Main ductwork shall be sized accordingly.

Table 8: Exhaust Duct Mains Sizing Calculation

- Exhaust Main Demand at Full Build, D: $D = (\text{TUM Peak Demand} \times 100\% \text{ Diversity}) + \text{Facilities Basebuild Requirements}$ - Exhaust System N capacity: $N = 115\text{-}120\% \times D$ - Exhaust Main Duct Sizing to include future units: $\text{Sizing} = N + 1F \text{ or } N+2F$ Exhaust load to be split evenly across Main Ductwork

- 8.3 **Ductwork Laterals/Submains Sizing Calculation** - Each Exhaust lateral/submain shall be sized so that the lateral's capacity meets 115-120% of the Exhaust demand at full build for the lateral (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, each Exhaust lateral/submain shall have 15-20% spare future capacity over the Exhaust demand at full build for the lateral/submain (after initial construction of the system). See **Table 9**.

Table 9: Exhaust Duct Laterals/Submain Sizing Calculation

- Exhaust Lateral/Submain Demand at Full Build (Tools + VMBs), $DL = \text{TUM Peak Demand for Lateral} \times 100\% \text{ Diversity}$ - Exhaust Lateral/Submain Duct Sizing = $115\text{-}120\% \times DL$
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- 8.4 **Duct Sizing Criteria** – Exhaust Main Duct Sizing shall be selected based on the full build capacity of the Exhaust system equipment, including the future equipment (N+1F or N+2F). It is recommended for the duct sizing to be designed with maximum velocities not to exceed the tables below: See **Table 10**.

Table 10 – Duct Sizing Criteria for Exhaust Ductwork

Duct Size	Maximum Velocity
10" to 18" (DN250 to DN450)	Less than maximum of 1,771 fpm (9.0 m/s)
20" to 26" (DN500 to DN 650)	Less than maximum of 1,968 fpm (10.0 m/s)
28" and larger (DN700 and larger)	Less than maximum of 2,363 fpm (12.0 m/s)

8.4.1 If the ductwork sizing is designed with velocities exceeding **Table 10**, the ductwork system pressure drop calculations shall demonstrate that the exhaust fan selections have adequate performance rating to meet the design parameters and system requirements.

8.5 **Duct Pressure Rating, Mains** – Exhaust ductwork mains shall be designed to meet the minimum pressure class of -15 inches w.g. (-3,750 Pa). If the system pressure at the main ductwork exceeds this amount, then the ductwork pressure rating shall meet or exceed the system pressure.

8.6 **Duct Pressure Rating, Submains/Laterals** – Exhaust ductwork laterals and submains shall be designed to meet a pressure class of -8 inches w.g. (-2,000 Pa). If the system pressure at the submain/lateral ductwork exceeds this amount, then the ductwork pressure rating shall meet or exceed the system pressure.

8.7 **Materials of Construction**

8.7.1 **Acid / Ammonia Exhaust Ductwork** – Ductwork shall be FM Approved Fluoropolymer-coated Stainless Steel, FM Approved Fiberglass Reinforced Plastic (FRP), or standard Fiberglass reinforced Plastic (requiring internal sprinklers) materials of construction.

- Fluoropolymer-coated Stainless Steel ductwork includes ETFE (Ethylene-Tetrafluoroethylene) and ECTFE (Ethylene-Chlorotrifluoroethylene) coating materials.
- It is recommended to use FM Approved ductwork materials, in lieu of standard FRP ductwork that is not approved by FM. Ductwork that is FM Approved per FM Standards 4922 and 4910 do not require internal sprinklers, per NFPA Standard 318 and FM requirements.

Note: Ductwork made of PVC, PP or PE shall NOT be allowed for sizes 6" and larger. The exception is smaller size PVC ductwork, sizes 4" (DN100) and below, which is typically used for tool hookup ductwork and VMB ductwork.

- **Gaskets, general:** In general, PTFE (Polytetrafluoroethylene, aka Teflon) tape gaskets or rope gaskets shall be used for Acid and Ammonia Exhaust. Gasket shall be installed on the duct flange lip surface.
- **Gaskets, special cases:** For special areas in which there is potential for liquid accumulation or condensation, such as WETs, Dry Etch, and Photo process areas, it is recommended to use Viton Full-Face type gaskets, 1/8" (3.0mm) thick. The Full-Face type gasket shall be installed

on the entire flange ring face of the angle rings, at each ductwork joint.

8.7.2 **General Exhaust Ductwork** – Ductwork shall be Galvanized Steel.

- **Gaskets:** In general, EPDM gaskets shall be used for General Exhaust.

8.7.3 **Solvent Exhaust Ductwork** – Ductwork shall be Stainless Steel or Teflon-coated Stainless Steel.

- **Gaskets, general:** In general, Teflon (PTFE) gaskets shall be used for Solvent Exhaust. Gasket shall be installed on the duct flange lip surface.
- **Gaskets, special cases:** For special areas in which there is potential for liquid accumulation or condensation, such as WETs, Dry Etch, and Photo process areas, it is recommended to use Viton Full-Face type gaskets, 1/8" (3.0mm) thick. The Full-Face type gasket shall be installed on the entire flange ring face at each ductwork joint.

8.7.4 **Scrubber Water Recirculation Piping** – piping shall be Schedule 80 PVC materials of construction.

8.7.5 **Chemical Dosing Piping** – piping shall be double contained piping. Interior tubing shall be PP, PE, or PFA. Exterior outer containment shall be Clear transparent PVC.

8.7.6 **Exhaust Ductwork Drain Piping** – piping shall be Schedule 80 PVC materials of construction.

8.8 Ductwork Installation

8.8.1 It is recommended to use round ductwork where possible. If this is not possible due to space constraints, rectangular ductwork is acceptable.

8.8.2 Ductwork branch lines and take-offs shall use increased area openings, such as conical or bell- mouth fittings (bull head tees shall not be allowed).

8.8.3 Ductwork takeoffs for branch ducts, spare dampers, POCs, etc. shall face directly sideways (with higher bottom surface elevation than the main) or upwards. Ductwork takeoffs shall not face downwards or diagonally downwards. The bottom surface elevation of the takeoffs shall not be at the same height or lower than the bottom surface elevation of the main. This is to prevent any liquid from entering or accumulating in the spare damper or lateral/submain from the main.

- 8.8.4 Inlet ductwork to Exhaust Scrubbers shall be designed so that any liquid or condensation in the exhaust ductwork will drain back into scrubber sump. The scrubber inlet elevation shall be at a lower height than the exhaust duct header elevation, to prevent any liquid from entering or accumulating in the exhaust duct header.
- 8.8.5 FRP ductwork located outdoors shall be painted for UV protection.
- 8.8.6 Ductwork shall be installed in accordance with SMACNA "Industrial Duct Construction Standards."
- 8.8.7 Prior to delivery to site from the factory, 100% of the Coated Stainless-Steel ductwork shall be factory spark tested to identify any flaws, defects, or damage of the duct coating.
- 8.8.8 Immediately prior to installation on-site, a minimum of 25% of the Coated Stainless-Steel ductwork shall be spark tested on-site to identify any flaws, defects, or damage of the duct coating.

8.9 Provisions for Main Expansion

- 8.9.1 At locations in which there is a planned future extension of the exhaust main ducts, headers, or laterals/sublaterals, provide full-size blastgates with capped blind flanges at the ductwork ends, to allow easier extension without interruption to the system operation.
- 8.9.2 At locations of unplanned future extension, provide full-size blind flanges at the ductwork ends.
- 8.9.4 The chemical distribution system shall be designed to allow for future increase of capacity and the addition of future scrubbers.

8.10 Spare Dampers for Future Laterals/Submains -

- 8.10.1 Additional spare dampers shall be included on the main ductwork, to allow installation of future laterals/submains for capacity expansion. Based on the number of laterals/submains included in the design, a minimum quantity of 10-15% of additional spare dampers for future laterals shall be included.
 - 8.10.2 Only Spare Dampers facing directly sideways (with higher bottom surface elevation than the main) or upwards shall be used. Spare Dampers shall not face downwards or diagonally downwards. The bottom surface elevation of the spare dampers shall not be at the same height or lower than the bottom surface elevation of the main. This is to prevent any liquid from entering or accumulating in the spare damper or lateral/submain from the main.
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8.11 Lateral Point-of-Connection Quantity and Sizes

- 8.11.1 The laterals shall have sufficient quantity of Point of Connections (POCs) to meet the tool layout and requirements.
- 8.11.2 A minimum quantity of 30-40% of additional spare POCs shall be provided on each lateral, for future tools.
- 8.11.3 Provide adequate spacing in between each POC, to allow proper airflow pattern and uniformity.
- 8.11.4 The sizes for each POC shall be checked and verified versus the actual tool requirements. Minimum POC size shall be DN250 (10”), but larger POC sizes can be provided, based on the actual tool requirements.
- 8.11.5 POCs shall be checked and verified versus the actual tool requirements. Minimum POC size shall be DN250 (10”), but larger POC sizes can be provided, based on the actual tool requirements.
- 8.11.6 Only POCs facing directly sideways (with higher bottom surface elevation than the lateral/submain) or facing upwards shall be used. POCs shall not face downwards or diagonally downwards. The bottom surface elevation of the POC takeoffs shall not be at the same height or lower than the bottom surface elevation of the lateral/submain. This is to prevent any liquid from entering or accumulating in the POC and tool hookup ductwork from the lateral/submain.

9.0 Duct Accessories Requirements:

9.1 Dampers and Blastgates

- 9.1.1 **Main Dampers at Laterals/Submains, FAB Level** – For Laterals/Submains in FAB Level, provide main balancing damper for the lateral outside of the FAB or at the level above the FAB, to allow balancing work to be performed without need to enter into FAB Cleanroom.
- 9.1.2 **Main Dampers at Laterals/Submains, Subfab Level** – For Laterals/Submains in Subfab Level, provide main balancing damper for the lateral in the Subfab level to allow balancing work to be performed directly in the same space.
- 9.1.3 Dampers on Mains and Laterals/Submains shall be heavy duty industrial, rectangular, multi-blade, opposed-blade type dampers. Damper operator shall be wormgear driver type.

- 9.1.4 Dampers for Lateral POCs shall be industrial duty, round butterfly type dampers with edge seals.
 - 9.1.5 All dampers shall be lockable type dampers to allow the damper setting to be set and fixed in-place.
 - 9.1.6 For motorized dampers with actuators, it is recommended to have square-shaped shafts connecting the damper and the actuator, to prevent slippage and mis-operation.
 - 9.1.7 **False Loading Dampers for Startup** - Spare dampers with mesh screens across the openings shall be provided on the main header and at the submain/laterals at Subfab level, to use for false loading during system startup. Upon system startup, open the spare dampers as required to provide sufficient false loading of the exhaust fans, for stable operation.
 - 9.1.8 **Blastgates** - Blastgates shall be provided where complete positive shutoff isolation is required. Examples include upstream of Scrubbers and Fans at the inlet connections, to prevent air leakage when taking equipment offline and isolating to perform PM activities.
 - For larger size blastgates, use blastgates with multiple smaller sized blades/gates and internal guide supports, to allow easier removal and insertion of the blastgates.
 - 9.2 **Duct Drains** – Low point drains shall be provided as to prevent any collection or accumulation of liquids or condensation in the ductwork.
 - 9.2.1 Provide low point drains at all low points, in addition to the locations specified in this section.
 - 9.2.2 Ductwork shall have slight slope towards one end or towards low point drains. Slope shall be approximately 1:500.
 - 9.2.3 **Ductwork Main Headers** – Provide duct drains at every 100 feet (30.5 meters) distance on the main header. There shall be minimum 3 low point drains per main header, one at each end of the and one at midpoint of the main header.
 - 9.2.4 **Ductwork Submains/Laterals** – Provide duct drains at every 100 feet (30.5m) distance on the laterals and submains. There shall be minimum 3 low point drains per submain/lateral, one at each end of the and one at midpoint of the lateral.
 - 9.2.5 **Scrubber Inlet** – Provide low point drain at ductwork inlet connection to Exhaust scrubbers.
 - 9.2.6 Low point drain and ductwork drain piping size shall be minimum DN40 (1-1/2 inch) diameter.

- 9.2.7 Ductwork drain piping shall connect to main piping via "Y-connection" type fittings, to allow for smooth drainage.
- 9.2.8 Scrubber and ductwork drain piping routed along the floor shall not obstruct or block any corridors or main walkways, for walking or cart access
- 9.2.9 For ductwork with in-duct sprinklers and at scrubber inlet/outlets, drains shall be sized and provided to accommodate the sprinkler water discharge. Typical ductwork drain size at sprinkler locations is DN100 (4") diameter.
- 9.2.10 Ductwork Drains shall be routed to a nearby wall, column, or Unistrut rack, and then connected to site drain with site glass assembly. Site glass assemblies shall be at an accessible location and shall consist of a 300mm (12-inch) high clear PVC sight glass and 2 isolation valves (above and below the sight glass).
 - Acid Exhaust shall be connected to a Site Acid Drain, such as IWW, HFW, CAW, etc.
 - Ammonia Exhaust shall be connected to a Site Ammonia Drain.
 - Solvent Exhaust shall be connected to a Site Solvent Drain, such as SW.
 - General Exhaust shall be connected to a Site Drain containing non-corrosives only, or the drain shall be terminated with a cap at a height of 1m (42 inches) above the floor level.
- 9.2.11 Locations of ductwork drains shall be determined prior to manufacturer's fabrication of the ductwork. Provide factory installed drain fittings and adapters in the ductwork. Avoid field-installed taps that may the compromise or affect the integrity of the ductwork, duct lining, etc.

9.3 Ductwork Test Ports, Inspection Ports, and Sampling Ports –
Ductwork test ports shall be provided to allow airflow measurements, pressure readings, and air sampling.

- 9.3.1 Locations of ductwork test ports and sampling ports shall be determined prior to manufacturer's fabrication of the ductwork. Provide factory installed drain fittings in the ductwork. Avoid field-installed taps that may the compromise or affect the integrity of the ductwork, duct lining, etc.

Test Ports/Traverse Ports

- 9.3.2 Main Ductwork - Each main shall have test ports at both ends of the main to allow airflow and pressure readings.
- 9.3.3 Lateral/Submain Ductwork - Each lateral/submain shall have test ports at both ends of the lateral/submain to allow airflow and pressure readings.
- 9.3.4 Exhaust Fans - Each exhaust fan shall have test ports upstream and downstream of the fan to allow airflow and pressure readings.
- 9.3.5 Exhaust Scrubbers - Each Scrubber shall have test ports upstream and downstream of the scrubber (at inlet and outlet) to allow airflow, pressure, and air quality readings.
- 9.3.6 Test ports shall be minimum 25mm (1 inch) diameter in size.

Inspection and Cleaning Ports

- 9.3.7 Inspection ports shall be provided on all exhaust ductwork at every 6 meter (20 feet) spacing intervals.
- 9.3.8 Inspection ports shall be minimum 50mm (2 inches) diameter in size, with threaded cap.
- 9.3.9 Exhaust Fans - Each exhaust fan shall have minimum 150mm (6 inches) cleaning port upstream and downstream of the fan to allow inspection and cleaning.

Emissions Sampling Ports

- 9.3.10 Two 100mm (4 inch) sample ports at 90 degrees shall be provided on the inlet and outlet of each scrubber to allow for monitoring of the scrubber's performance, and for testing by regulatory agencies.
- 9.3.11 Two 100mm (4 inch) sample ports at 90 degrees shall be provided on all exhaust stacks to allow for monitoring and testing by the regulatory agency.

9.4 Fire Protection Sprinklers

- 9.4.1 Sprinkler heads shall be provided inside the exhaust ductwork, for ductwork constructed of combustible materials, sizes 10 inches (254 mm) and greater in diameter, or with cross section area equal or greater to 80 inches² (516 cm²). This is per NFPA 318 Section 4.1.2.6 and FM Standard 7-78.

- Internal fire sprinklers are not required for FM Approved ductwork.
 - Therefore, the internal sprinkler requirement applies to Acid and Ammonia Exhaust ductwork constructed of FRP materials of construction, that is non-FM approved.
- 9.4.2 Sprinkler heads shall be provided on the inlet and outlet connection of each Exhaust Scrubbers. The sprinkler head can be located on either the inlet and outlet ductwork, or on the scrubber body inlet and outlet connections.
- 9.4.3 If required by local code, provide internal sprinkler heads inside ductwork on both sides of fire rated partitions.
- 9.4.4 Sprinkler heads installed inside Acid and Ammonia exhaust ductwork shall be corrosion resistant and/or protected from corrosion. Allowable types include FM approved wax, lead, or wax-over-lead coated sprinkler heads, or plastic-coated sprinkler heads (coated with epoxy, TFE resin, etc.). Additional double layer polypropylene or polyethylene bag may be used to contain the sprinkler head.
- 9.4.5 Sprinkler heads shall be connected to the sprinkler branch piping with flexible hose assembly.
- 9.4.6 Provide inspection ports in the ductwork at the location of each sprinkler head, to allow visual inspection of the sprinkler heads.
- 9.4.7 Provided adequate low point drains in the exhaust ductwork with sprinkler heads. Drains shall be sized adequately to be capable of draining all sprinkler water discharged, in the event that sprinklers are activated.
- 9.4.8 Locations of sprinkler heads shall be determined prior to manufacturer's fabrication of the ductwork. Provide factory installed sprinkler adapter fittings in the ductwork for each sprinkler head. Avoid field-installed taps that may the compromise or affect the integrity of the ductwork, duct lining, etc.

9.5 Fire Dampers

- 9.5.1 Fire Dampers shall not be installed in Exhaust ducts, unless required by local code or jurisdiction.

Fire dampers inside Exhaust ducts are prohibited by NFPA Standard 318 Section 5.3.5, and International Mechanical Code Section 510.6.1

Note: If the local code or jurisdiction requires fire dampers in the exhaust ductwork, it shall take precedence and the fire dampers shall be installed.

9.6 **Other Accessories** - Provide other accessories as outlined in **Table 11**.**Table 11: Accessories to be installed**

Accessories	Type & Installation Location
Piping Insulation	- For site geographic locations that experience freezing temperatures, scrubbers and their associated piping and components that are located outdoors shall be provided with a means of preventing freezing. - For site geographic locations that experience freezing temperatures, with scrubbers located outdoors, provide piping insulation and piping Heat Trace to prevent freezing
Ductwork Supports	- Provide duct hanger and support material gauges in compliance with details on the Drawings. Where details are not provided, comply with SMACNA Industrial Duct Construction Standards. - Ductwork hanger and suspension material may be threaded rod, structural shapes, or strut material. The use of wire as suspension material is prohibited. - Ductwork hanger supports and spacing shall take into account the potential weight of water when the ductwork contains fire sprinklers. - Refer to the specific project's Construction Specification, (i.e. Spec Section 15140 or equivalent Spec) for additional requirements.
Labeling and Identification	- Provide identification labels at minimum of every 6m distance and change of direction - Refer to the specific project's Construction Specification, (i.e. Spec Section 15190 or equivalent Spec) for additional requirements. <u>Ductwork</u> - Labels on Exhaust Ductwork shall be Yellow Background with Black Lettering "AE" nomenclature shall be used for Acid Exhaust "AME" nomenclature shall be used for Ammonia Exhaust (also Alkali Exhaust) "SE" nomenclature shall be used for Solvent Exhaust (also known as VOC Exhaust) "GE" nomenclature shall be used for General Exhaust

	Piping • Labels on Chemical Dosing Piping shall be Yellow Background with Black Lettering • Labels on Exhaust Drain Piping shall be Yellow Background with Black Lettering
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10.0 Sustainability Design:

10.1 **Exhaust Fan Parallel Header Design** – Exhaust Fans shall be arranged in parallel and connected to a common ductwork header upstream of the fans and downstream of the Scrubbers/Abatement Units. This allows multiple exhaust fans to operate together in parallel, to lower the fan operating speed and fan energy consumption.

10.2 **Ductwork Pressure Drop Sizing** – It is recommended to increase the ductwork sizes where possible, to minimize the air velocity and pressure drop through the ductwork. This will result in lower exhaust fan energy consumption.

10.3 **Exhaust Load Reduction** – It is recommended to reduce the exhaust load as much as possible, including during tool hookup design, exhaust balancing, and system operation. This will result in lower energy consumption of both the Process Exhaust System, and the Cleanroom Makeup Air Handling Systems.

10.3.1 General Exhaust Reduction – Heat Exhaust from certain Implant and Metrology tools can be discharged directly into Cleanroom space, instead of being connected to General Exhaust System. When implementing this, the loading of the Cleanroom Dry Cooling coils shall be checked to ensure there is adequate cooling capacity for the additional heat load.

11.0 Electrical Design:

11.1 **Power Segregation** – Electrical power supply for exhaust equipment including scrubbers, fans, and VOC Abatement Units shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.

11.2 **Un-interrupted Power for Controls** – 100% of exhaust system controls including PLCs, LCPs, and other auxiliary control power shall be connected to an Un-interrupted Power source to allow continuous operation of the exhaust system during momentary power interruption or voltage sags, spikes, and fluctuation events.

- 11.3 **Emergency Power – Exhaust Fans** – A minimum of 50% of the exhaust fans for each system shall be connected to Emergency Power.

This is per requirements of Factory Mutual (FM) Standard 1056, National Fire Protection Agency (NFPA) Standard 318, and International Building Code (IBC) Section 415, to maintain 50% capacity (exhaust airflow) upon a power loss.

The interpretation is that by having 50% of the exhaust fans on emergency power, it is sufficient to provide approximately ~50% capacity (exhaust airflow) upon a power loss, to satisfy the requirements.

To maintain proper cleanroom pressurization levels, then a minimum 50% of Cleanroom Makeup Air Units shall also be on emergency power. This requirement will be covered in a separate Global System Standard document for Makeup Air Units.

- 11.4 **Emergency Power – Scrubber Pumps** - A minimum of 50% of the Scrubbers and their associated recirculation pumps shall be connected to Emergency Power.

- 11.5 **Variable Frequency Drives** – All fans shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual fan motor size.

11.5.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

11.5.2 Variable Frequency Drives shall be configured to “Fail Last Speed” configuration upon any failure, including but not limited to, loss of power, loss of communications, etc.

11.6 **Motors**

11.6.1 Exhaust fan motors shall be inverter duty rated with insulated bearings and shaft grounding kits. Motors shall be compatible with the associated VFD.

11.6.2 All motors located outdoors shall be IP56 rated protection.

11.6.3 All motors shall be National Electrical Manufacturers Association (NEMA) IE3 rated or better.

- 11.7 **Local Disconnect Switches** – Each exhaust fan shall have a lockable type, local disconnect switch installed adjacent to the fan, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the fan.

11.7.1 All disconnect switches located outdoors shall be IP56 rated protection.

- 11.8 **Lightning Protection** – Provide lightning protection for all outdoor equipment, including the exhaust discharge stacks.
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12.0 Instrumentation:

- 12.1 The following Instrumentation shall be installed and monitored/controlled via FMCS: See **Table 12**.

Table 12 – Instrumentation Devices

Accessories	Type & Installation Location
Pressure Transmitters	• Acid and Ammonia Exhaust: Exhaust Duct Main header upstream of the scrubbers shall have a minimum of three pressure sensing locations – one on each end, and one at the center. Each pressure sensing location shall have two pressure transmitters, one duty and one redundant backup (for N+1). • Solvent Exhaust: Exhaust Duct Main header upstream of the VOC Abatement Units shall have a minimum of three pressure sensing locations – one on each end, and one at the center. Each pressure sensing location shall have two pressure transmitters, one duty and one redundant backup (for N+1). • General Exhaust: Exhaust Duct Main header upstream of the General Exhaust Fans shall have a minimum of three pressure sensing locations – one on each end, and one at the center. Each pressure sensing location shall have two pressure transmitters, one duty and one redundant backup (for N+1). • Install pressure transmitters at Lateral end-of-line for a minimum of 4 laterals per system, in the critical zones (i.e. Photo, Diffusion, and WETs process areas) to allow online monitoring. Additional pressure transmitters can be installed at other laterals for additional monitoring.
Exhaust Flowmeters	• Install a flowmeter on each exhaust fan and/or scrubber/VOC Abatement Unit to allow continuous exhaust load monitoring.
Scrubber Integral Transmitters	• Integral Differential Pressure Transmitter shall be provided with each scrubber (across the packing and or packing + mist eliminator). • Integral Flow Transmitter shall be provided with each scrubber, to indicate the water recirculation flowrate.

	• Integral pH Transmitter shall be provided with each scrubber • Integral Conductivity Transmitter shall be provided with each scrubber • Integral Sump water level sensors shall be provided with each scrubber
Leak Sensors/ Transmitters	• Scrubber Double Containment Sump shall have Leak Sensor/Transmitter • Chemical Dosing Central Tank Containment Sump shall have Leak Sensor/Transmitter • Chemical Dosing Localized Day Tank Containment Sump shall have Leak Sensor/Transmitter • Double-contained tubing for chemical dosing supply shall have leak sensors installed in the lowest elevation point of the chemical distribution pipe routing. The leak sensor shall be located between the inner tubing and the outer transparent PVC piping.
Additional Spare Ports	• All exhaust laterals shall have a spare port and installed at the lateral end-of-line, to allow for future pressure transmitters to be installed. The future pressure transmitters will allow continuous SCADA monitoring of each lateral pressure.

12.2 Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.

12.3 The Pressure Differential Transmitters on exhaust ductwork shall reference the Zero Reference Pipe (ZRP). This shall be the same ZRP used for Cleanroom pressurization reference.

12.3.1 ZRP shall reference outdoor atmospheric pressure. ZRP external probe shall be located in a place with minimum wind interference, and shall have adequate cover protection from wind.

13.0 Control Sequence Philosophy:

- 13.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 13**.

Table 13: System Control Logic

Control Parameter	Control Logic
Exhaust Pressure Control	- Speed output of Exhaust Fan VSDs shall be modulated to maintain the pressure setpoint at the Exhaust Main duct. - The pressure setpoint shall be averaged across the number of pressure sensing locations spread across the Exhaust Main duct.
Scrubber pH Control	- Scrubber pH control shall be automatically done via the scrubber's integral control algorithm via the Local Control Panel. - Acid Scrubber pH shall be maintained at setpoint of 8-9 (adjustable). - Ammonia Scrubber pH shall be maintained at setpoint of 3-4 (adjustable). - Upon a call for chemical dosing at the Scrubber, the chemical dosing control valve shall open to supply chemical to the Scrubber and adjust the pH towards the setpoint.
Scrubber Blowdown and Makeup Water Control	- Scrubber conductivity blowdown and makeup water control shall be automatically done via the scrubber's integral control algorithm via the Local Control Panel. - Upon sensing high conductivity levels, the scrubber blowdown drain valve shall open to begin blowdown. - Upon sensing low level, the scrubber makeup water valve shall open to begin makeup.
VOC Abatement Unit Control	- VOC Abatement Unit control shall be automatically done via the unit's integral control algorithm via the Local Control Panel, including but not limited to the following: - Damper Controls - Unit-mounted Fan Controls - Zeolite Rotor controls - Burner Controls - Desorption Temperature Control - Oxidizer Temperature Control
Exhaust Fan Start/Stop	Exhaust Fan Start/Stop shall be done automatically via SCADA based on required number of running fans

13.2 A detailed Sequence of Operation (SOO) shall be developed specifically for each system, as part of each construction project.

13.3 Control Valves and Dampers

13.3.1 The following control valve and damper fail positions shall be configured. See **Table 14**.

Table 14: Control Valve and Damper Fail Positions

Control Valve or Damper	Control Valve or Damper Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
Control Damper at Scrubber inlet	Fail Last State	Fail Last State	Fail Last State
Control Damper at Fan inlet	Fail Last State	Fail Last State	Fail Last State
Scrubber pH Control Chemical Dosing Control Valve	Fail Closed	Fail Closed	Fail Closed
Scrubber Makeup Water Control Valve	Fail Closed	Fail Closed	Fail Closed
Scrubber Blowdown Drain Control Valve	Fail Closed	Fail Closed	Fail Closed
VOC Unit Inlet and Outlet Control Dampers	Fail Last State	Fail Last State	Fail Last State
VOC Main Header Bypass Control Damper	Fail Last State	Fail Last State	Fail Last State

13.3.2 The control valves and dampers shall be double-acting type actuators, capable of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

13.3.3 After a loss event of control air, power, or PLC signal, and after it is restored, the control valve or damper output command/position shall be set to the same command/position it was operating at prior to the loss event.

- 13.3.4 All control valves and dampers shall have a manual override bypass, to allow manual operation of the valve or damper upon a failure.
- 13.3.5 For motorized dampers with actuators, it is recommended to have square-shaped shafts connecting the damper and the actuator, to prevent slippage and mis-operation. Prior to system startup, perform inspection and operations check of all dampers and actuators including shaft, linkages, and connection points.

14.0 Smart Facilities Design:

The following are additional enhancements that can be considered and included to enable Smart Facilities Design and improved monitoring/control of the Exhaust systems and equipment.

- 14.1 **Lateral Pressure Online Monitoring** – Consider installing pressure transmitters at all exhaust laterals at each end-of-line location, to allow continuous SCADA monitoring and alarm of each lateral pressure.
- 14.2 **Load Demand Online Monitoring** – Install flowmeters at the main ductwork headers and at each scrubber/fan, to allow continuous SCADA monitoring and tracking of exhaust load.
- 14.3 **Fan Vibration Online Monitoring** – Consider installing vibration sensors on each fan and motor, to allow continuous monitoring and alarm of fan vibration levels.
- 14.4 **Emissions Online Monitoring** – Consider installing online monitoring of Solvent Exhaust emissions levels and DRE efficiency of VOC Abatement Units.

15.0 Inter-Discipline Coordination Matrix:

- 15.1 The following provisions shall be provided by each of the following disciplines, and coordinated accordingly: See **Table 15**.

Table 15: Inter-Discipline Coordination Matrix

Discipline	Provision
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Architectural	• Provide mechanical room with appropriate fire rated walls and partitions for scrubbers, exhaust fans, etc. • If located indoors, scrubbers and exhaust fans shall be provided with a concrete housekeeping pad. • Chemical Dosing units shall be provided with concrete housekeeping pads. • Scrubbers and Chemical Dosing Units shall be provided with secondary containment. • Allocate stack location on roof for exhaust discharge. • Cut openings in walls and floors for ducting penetrations, and seal openings after duct installation • Provide adequate fire stopping at all required fire wall penetrations • Provide access platforms, ladders, and stairways for Exhaust Scrubbers as needed to perform maintenance and repair/replacement on scrubber components including packing, spray nozzles, and mist eliminator. • Provide access platforms, ladders, and stairways for Exhaust Stacks as required to perform regular maintenance and emissions testing at sampling port locations
Mechanical	• Provide makeup water piping and accessories as outlined in this Standard • Provide natural gas piping and accessories as outlined in this Standard • Provide fire sprinkler piping, sprinkler heads, and accessories as outlined in this Standard • Provide drainage and accessories as outlined in this Standard
Electrical	• Provide VFDs for all exhaust fan motors. • Provide local disconnect switches for all exhaust fan motors and pump motors. • Provide power wiring for all Electrical equipment/ components • Provide grounding for all Electrical equipment/ components • Provide lightning protection for outdoor equipment including exhaust discharge stacks. • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Water / Wastewater	• Provide makeup water source for the Scrubber. Source of makeup water for scrubbers can be Non-potable water (NPW), Condensate Drain (CD), or other Reuse/Reclaimed Water (RW). • Provide drain source for the Scrubber Drain water piping • Provide drain source for the Ductwork Drain water piping • Provide NaOH and H2SO4 bulk chemical feed for the main Central Chemical Dosing system

	• Install NPW hose bibs with caps near the exhaust fans, to be used for cleaning/washing the fans.
Controls	• Provide control damper and control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide Zero Reference Pipe (ZRP) for atmospheric pressure reference of Exhaust and Makeup Air Systems • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, etc. • Provide FMCS/SCADA system connectivity of all Local Control Panels

16.0 Quality Inspection and Testing:

16.1 **Inspection and Test Plan (ITP)** – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

16.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Assembly Procedures, Qualifications, and Inspection
- Piping and Duct Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Ductwork Coating Spark Testing - Immediately prior to installation on-site, a minimum of 25% of the Coated Stainless-Steel ductwork shall be spark tested on-site to identify any flaws, defects, or damage of the duct coating.
- Ductwork Leak Testing – A minimum of 25% of Process Exhaust ductwork shall be leak tested.
- Leak Testing and Pressure Testing
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist

- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.

16.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

17.0 Startup Commissioning / Site Acceptance Test:

17.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Testing & Commissioning / Acceptance Tests.

17.1.1 Fans

- Fan Motor Rotation Check
- Fan/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Fan Standalone Startup Procedure per Manufacturer's procedure
- Fan Vibration Level Checks
- Verify Fans Operation through its operating range (0 to 50/60 Hz)
- Verify Fan Start/Stop Function to Rotate Operating Fan
- Verify Fan Auto-Start sequencing, upon failure of an operating Fan
- Verify Fan Speed control in response to Exhaust pressure fluctuations

- Verify Fan Inlet Motorized Damper opens upon fan startup and closes upon fan stop, while keeping exhaust pressure stable
- Confirm Exhaust system pressure control is stable during fan manual start/stop rotation
- Confirm Exhaust system pressure control is stable during fan fault/trip event and next fan auto-start sequencing
- Verify VSD fails last speed upon any failure
- Verify control interface and communication checks to FMCS/SCADA

17.1.2 Scrubbers

- Pump Motor Rotation Check
- Pump/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- Control interface and communication checks to Local Control Panel, PLC, and FMCS/SCADA
- Scrubber Standalone Startup Procedure per Manufacturer's procedure
- Verify Recirculation Pump Start/Stop Function to Rotate Operating Recirculation Pump
- Verify Standby Recirculation Pump Auto-Start sequencing, upon failure of an operating Recirculation Pump
- Verify Scrubber Inlet Motorized Damper opens upon scrubber startup and closes upon scrubber stop, while keeping exhaust pressure stable
- Confirm Exhaust system pressure control is stable during scrubber start/stop rotation
- Verify control interface and communication checks to Local Control Panel and FMCS/SCADA
- Perform independent 3rd party Scrubber performance analysis on abatement removal efficiency, after Exhaust system is loaded (requires manufacturing tools to be in operation to provide sufficient chemical loading)

17.1.3 VOC Abatement Units

- All Fan Motor Rotation Checks
- All Fan/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- Rotor Motor Rotation Checks
- Rotor Alignment Check and Lubrication

- VSD Settings and Functionality Checks
- Control interface and communication checks to Local Control Panel, PLC, and FMCS/SCADA
- VOC Abatement Unit Standalone Startup Procedure per Manufacturer's procedure
- Desorption and Oxidizer Temperature Checks
- Verify VOC Abatement Unit Inlet and Outlet Motorized Damper opens upon unit startup and closes upon unit stop, while keeping exhaust pressure stable
- Confirm VOC Bypass Damper opens upon unit shutdown/fault and closes upon unit startup, while keeping exhaust pressure stable
- Confirm Exhaust system pressure control is stable during VOC Abatement Unit start/stop rotation
- Confirm Exhaust system pressure control is stable during VOC Bypass Damper operation
- Control interface and communication checks to Local Control Panel FMCS/SCADA
- Perform independent 3rd party VOC Abatement Unit performance analysis on abatement and destruction removal efficiency (DRE), after Exhaust system is loaded (requires manufacturing tools to be in operation to provide sufficient chemical loading)

17.1.4 Chemical Dosing Units

- Pump Motor Rotation Check
- Pump/Motor Alignment Check and Lubrication
- Confirm central storage tank and localized daytank begins filling operation upon sensing low tank levels, and stops filling operation upon sensing high tank levels
- Verify Standby Chemical Dosing Pump Auto-Start sequencing, upon failure of the operating Chemical Dosing Pump
- Leak Sensor Checks
- Control interface and communication checks to Local Control Panel and FMCS/SCADA

17.1.5 Integrated system performance test

- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- For VOC Abatement Units, confirm Auto-switch to backup temperature transmitter(s) upon failure of main temperature transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.

- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test

18.0 Reference Specifications:

18.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 16**.

Table 16 – Specification Sections

Spec Section	Spec Contents
Section 15060	Pipe and Pipe Fittings
Section 15190	PVC and CPVC Piping Systems
Section 15101	Valves
Section 15120	Piping Specialties
Section 15121	Process piping Specialties
Section 15130	Indicating Devices
Section 15140	Supports and Anchors
Section 15190	Mechanical Identification
Section 15240	Mechanical Sound and Vibration Control
Section 15260	Piping Insulation
Section 15280	Equipment Insulation
Section 15330	Wet Pipe Sprinkler Systems
Section 15461	Process Exhaust System
Section 15515	Hydronic Specialties
Section 15540	Pumps
Section 15860	Exhaust Fans
Section 15861	FRP Exhaust Fans and Stacks
Section 15891	Metal HVAC Ductwork and Accessories
Section 15892	Metal Industrial Ductwork and Accessories
Section 15993	Air Systems Testing, Adjusting, and Balancing
Section 15170	Motors
Section 15171	Variable Frequency Drives

19.0 Document Control

19.1 Approval: GLOBAL_FAC_GFTT_APPR

19.2 Notification:

GLOBAL_FAC_MANAGERS
 GLOBAL_FAC_COMMUNICATIONS
 GLOBAL_FAC_CONSTRUCTION
 GLOBAL_FAC_NOTIFY
 FCG_F2_FAC_NOTIFY
 FCG_F4_FAC_NOTIFY
 FCG_F6_FAC_NOTIFY
 FCG_F10W_FAC_NOTIFY
 FCG_F10N_FAC_NOTIFY
 FCG_MTTW_FAC_NOTIFY
 FCG_F15_FAC_NOTIFY
 FCG_F16_FAC_NOTIFY
 FCG_MMP_FAC_NOTIFY
 FCG_MMY_FAC_NOTIFY
 FCG_MSB_FAC_NOTIFY
 FCG_MXA_FAC_NOTIFY
 MTB_FAC_ALL_NOTIFY

19.3 Final Viewer Access

All Micron Team Members at all Sites

19.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#).

19.5 Review

Biennially (two years)

20.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	08/25/20